KTH ROYAL INSTITUTE OF TECHNOLOGY

Introduction to IL2234 Digital Design using HDL

Electrical Engineering and Computer Science (EECS)

Period 1 – 2025

Course Structure

- Lectures
  - Digital design concepts
  - RTL design methodology
  - Hardware modeling using HDLs
  - Verification concepts
- Quizzes
  - To gauge the class's level in preparation for the upcoming lectures
- Exercises
  - Practice on the concepts we learn in the lectures
  - Technical requirements
    - > *Simulation & synthesis tools*
    - > *Development Boards*

- Homework
  - Individual work
  - Exercise on the concepts
  - Submit solutions online
- Project
  - Work in teams
  - Lab Sessions for Q&A and Project Delivery
- Exams
  - 1 digital exam (online or in class)
  - 1 oral exam

Lectures Content

- Functionality
- Gate-level
- Cascading
- SystemVerilog
- Application

- Topic 1: Introduction
- Topic 2: Combinational Components and their modelling using HDL
- Topic 3: Sequential Components and their modelling using HDL
- Topic 4: Controllers and their modelling using HDL
- Topic 5: Memory and buses
- Topic 6: RTL Design Methodology and Hierarchical Modelling
- Topic 7: Verification Concepts

Exercise Sessions

- Simulation tool & Simple modeling &Testbench
- A combinational circuit
- Synthesis tool & FPGA
- A sequential circuit
- Verification examples
- Sample Exam

Homework

- 4 Homework is to be done as individual work
  - 2 Practice assignments (P/F)
    - > *You need to solve at least one of the problems.*
    - > *You need to submit a solution that will be peer reviewed*
    - > *Solutions will be published after the peer review*
  - 2 Graded assignments
    - > *Each assignment gives 10 points*
    - > *You need at least 4 in each assignment*
    - > *We might call you to explain your solution*

Project

- Milestone 1 – ALU Design and Modelling
- Milestone 2 – Register File Design, Modeling, Synthesis, and FPGA Implementation
- Milestone 3 – Controller Design and Modeling
- Milestone 4 – Microprocessor Simulation, Synthesis, and FPGA Implementation

Grading

- Grade A-F
  - *Labs (P/F) → You need to complete all Labs to pass*
  - *Exam (P/F) → You require 65% of the points to pass the digital exam*
  - *Homework 3 and 4 → At least 4 points in each, look at the table for more details*
  - *Oral exam*

Grades	Requirements
Grade E	Labs (P), Exam (P), HW (8)
Grade D	Labs (P), Exam (P), HW (12)
Grade C	Labs (P), Exam (P), HW (16)
Grade B	Labs (P), Exam (P), HW (18), Oral Exam
Grade A	Labs (P), Exam (P), HW (18), Oral Exam

Preliminary Course Timeline

Week	Day	Lectures	Homework	Project & Labs	Exercise
	Tuesday	0_Syllubes & Topic1_Introduction			
	Wednesday	Topic 2_Combinatorial Components			
Aug 25-31	Thursday	Topic 2_Combinatorial Components			
	Monday				Simulation tool & Simple modelling &TB
	Tuesday	Topic 2_Combinatorial Components			
Sep 1-7	Wednesday	Topic 2b_Hardware modelling using HDLs			
	Tuesday	Topic 3_Sequential Components	HW 1 - Combinational	Milestone 1 - ALU	
	Wednesday				A combinational circuit
	Thursday	Topic 3_Sequential Components			
Sep 8-14	Friday	Topic 4_Controllers			
	Monday	Topic 4_Controllers	HW2-Sequential	Milestone 2 - RF & FPGA	
	Tuesday			Lab1	
	Wednesday				Synthesis tool & FPGA
Sep15-21	Thursday	Topic 5_Memory and buses			
	Tuesday	Topic 6_RTL Design Methodology		Milestone 3 - Controller	
	Wednesday				A sequential circuit
	Thursday			Lab2	
Sep 22-28	Tuesday	Topic 6_RTL Design Methodology	HW3-FSM&RTL Design		
	Monday	Topic 6_RTL Design Methodology			
	Tuesday	Topic 7_Verification			
Sep 29-5	Thursday	Topic 7_Verification		Milestone 4 - Microprocessor	Sample Exam
	Monday			Lab3	
	Tuesday	Topic 7_Verification	HW4 - Bus & Assertions		
Oct 6-12	Thursday				
Oct 13-19	Monday			Lab4	

Course Approach

- All course materials, including lectures, homework, and projects, will be posted on Canvas
- Lectures will be uploaded to Canvas after the class
- All instructions for doing homework and projects will be available on Canvas
- Canvas is constantly updated, so please always check for the latest information
- You can discuss with each other, ask questions, etc., in the discussion forum on Canavs

The Scope of this course

What you should know

- Basic CMOS
  - Circuit Analysis
  - CMOS
  - Gates
- Basic Digital Design
  - Boolean Algebra
  - Number Representation
  - Binary Arithmetic
  - Logic Minimization

Digital Design with HDL

What you will learn in this course

- **Advanced Digital Design**
  - Combinatorial Components
  - Sequential Components
  - FSMs
  - Data Path & Controller
  - Memories & Busses
  - RTL Design Methodology
  - Testing and Verification
- SystemVerilog
  - Modelling
  - Testbenches
  - Synthesis Aspects

Complementary Courses: SOC Design | Computer Architecture | Embedded Systems

Course Contacts

- Examiner
  - Ahmed Hemani hemani@kth.se
- Lectures
  - Nooshin Nosrati nosrati@kth.se
- Exercises
  - Jordi Altayó jordiag@kth.se
- Homework
  - Saba Yousefzadeh sabay@kth.se
- Labs
  - Saba Yousefzadeh sabay@kth.se
  - Jordi Altayó jordiag@kth.se

Literature

- Taraate, V. SystemVerilog for Hardware Description RTL Design and Verification.
- Mehta, A. B. Introduction to SystemVerilog.
- Sutherland, S., Davidmann, S., & Flake, P. SystemVerilog for Design: A Guide to Using SystemVerilog for Hardware Design and Modeling.
- Spear, C., & Tumbush, G. SystemVerilog for Verification A Guide to Learning the Testbench Language Features
- Cerny, E., Dudani, S., Havlicek, J., & Korchemny, D. SVA: The Power of Assertions in SystemVerilog
- Brown, S. D., & Vranesic, Z. G. Fundamentals of digital logic with VHDL design

Online resources

- https://www.chipverify.com/
- https://www.verilogpro.com/
- https://www.doulos.com/
- https://vlsiverify.com/
- https://verificationguide.com/
- https://edaplayground.com/